

Create a professional single-page engineering web application (HTML, CSS, JavaScript in one file) named: "Microstrip Patch Antenna Design Calculator"
Purpose: Design and visualize a rectangular inset-fed microstrip patch antenna for HFSS/CST users. Theme: - Dark RF engineering theme - Responsive layout - Print-friendly report mode - Professional laboratory appearance
INPUTS 1. Operating Frequency f (GHz) 2. Substrate Type - FR4 - Rogers RT5880 - Custom If FR4 selected: $\epsilon_r = 4.4$ $h = 1.6$ mm

Show more

Check frontend design skill >

Now I'll build this comprehensive engineering application. This is a large, complex file — let me build it carefully.

Ran 4 commands, created a file >

All checks pass. Let me present the file.

Presented file >

Here's your complete **Microstrip Patch Antenna Design Calculator** — a single 2,300-line self-contained HTML file (~94 KB). Here's what's included:

① Design & Calculate Tab

- Real-time calculations triggered on every keystroke/change
- Substrate selector: FR4, Rogers RT/duroid 5880, or fully Custom (ϵ_r , h , $\tan \delta$)
- Conductor selector: Copper/Silver/Gold/Aluminum/Custom (with σ entry)
- Copper thickness with 0.035 mm default
- **18 output result cards** grouped by: Patch, Ground, Feed/Inset, Material Stack, Simulation Box & Mesh
- Live status bar showing f , substrate type, ϵ_r , λ_0
- λ -strip showing λ_0 , λ_g , $\lambda/4$, $\lambda/20$ at all times

② Engineering Diagrams Tab

Three SVG views with auto-updating dimensions and engineering arrows:

- **Top View** — patch, inset notches, feed, gap, waveguide port, 7 annotated dimension lines
- **Side View** — cross-section with exaggerated vertical scale showing all layers (air, conductor, substrate, ground)
- **Isometric 3D** — projected view with labelled patch, feed, substrate, wave port plane, and axis triad

③ Formula Reference — 14 complete equations

Hammerstad fringing, Bahl & Trivedi 50Ω feed, inset impedance, skin depth, mesh sizing, and more

④ Parameter Guide — 17-row table

Parameter | Meaning | Effect on antenna performance

⑤ HFSS / CST Setup Tab

Six recommendation cards: Wave Port, Air Box/Boundary, Mesh Strategy, Frequency Sweep, Material Assignments, Post-Processing — with live-updating code values

📄 Printable Report

Click **Print Report** to get a full white-background report with input table, calculated results table, simulation parameters, formula summary, design notes (including expected gain/bandwidth/efficiency), and date/time stamp — ready for **Ctrl+P** to PDF.



Microstrip patch antenna calculator
Code · HTML

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